

Dr. Lorenz Ennio Gärtner

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Technical Expertise

Statistics & ML	Bayesian and frequentist inference , MCMC posterior sampling (PyMC), profile likelihoods, constrained optimisation, uncertainty quantification, supervised learning, automatic differentiation
Programming	Python (NumPy , SciPy , JAX , PyTorch), C++, basic Julia, $\LaTeX$, Mathematica
Tools	Git, Docker, CI/CD (GitHub Actions), pytest, packaging (PyPI, conda), HPC, profiling & optimisation, open-source maintenance , agentic development workflows
Languages	German (native), English (fluent); French, Italian (basic)

Work & Research Experience

Postdoctoral Researcher – *Democratizing statistical models* 2026 – present
BMFTR ErUM-Data project *Ludwig Maximilians University, Munich, Germany*

- **Composable models:** Statistical methods and open-source tooling for probabilistic models that can be shared and combined across domains, with propagated uncertainties.
- **Verifiability:** Results from one experiment can be checked and reused by another, without re-running the original analysis.

Scientific Advisor to MP Lars Rohwer Mar. – May 2026
German Parliament (Bundestag) *Berlin, Germany*

- **Technical consulting:** Advised a Member of the German Bundestag on energy, economic policy, and emerging technologies, turning ambiguous topics into clear recommendations.
- **Communication & coordination:** Translated frontier research into briefings, speeches, and media outputs while aligning political and scientific stakeholders.

PhD Researcher – *Accelerating Discovery in Particle Physics* 2022 – 2025
Belle II experiment *Ludwig Maximilians University, Munich, Germany*

- **Method:** Developed a novel statistical method for reinterpreting published measurements [3] ([publications](#)).
- **Software:** Built the open-source Python package [redist](#), now applied within the Belle II [2] and LHCb [4] collaboration.
- **Application:** Tested light invisible resonances in $B^+ \rightarrow K^+ X$ with the resulting public likelihood, finding a 3.0σ preference over the Standard Model [1].
- **Open source:** Contributed to [pyhf](#), [cabinetry](#), and [eos](#): declarative model specification, inference, and theory predictions.
- **Strategy:** Input to the European Strategy for Particle Physics Update 2026, via the Forum for Open Science [5].
- **Communication:** Lectured on Bayesian and frequentist inference at the [Inverted CERN School of Computing](#). 10+ international conference talks ([ICHEP](#), [PyHEP](#), [WLCG/HSF](#)). Designed a scientific-Python course for 60+ students; [tutorials](#) for 50+ researchers.

Applied Researcher & Policy Coordinator 2021 – 2022
International Atomic Energy Agency, Fast Reactors Team, Internship *Vienna, Austria*

- **Project coordination:** Coordinated a cross-disciplinary IAEA project connecting fusion and fission experts, organising workshops for 50+ experts.
- **Technical writing:** Authored and reviewed scientific proposals, papers, and presentations.

- **Applied machine learning:** Improved Belle II particle classification accuracy by >10% with a neural network replacing expensive detector simulations.
- **Large-scale simulation:** Evaluated observational astrophysics data against theoretical models using C++ Athena++ simulations.
- **Scientific tooling:** Extended the [redMaPPer](#) galaxy cluster finder with uncertainty estimation and masking corrections.

Education

PhD Physics: Particle Physics

2022 – 2025

Accelerating discovery: model-agnostic likelihoods for the reinterpretation of particle physics results *summa cum laude (best possible)*

Ludwig Maximilians University, Munich, Germany

MSc Physics: Theoretical & Computational Physics

2018 – 2020

*Very good, 1.09**Ludwig Maximilians University, Munich, Germany*

BSc Physics with Theoretical Physics

2015 – 2018

*First Class, 86.7% (1st in cohort)**University of Manchester, Manchester, United Kingdom*

Professional Development

2024 – 2026

- [BlueDot AGI Strategy Course](#) (2026): technical trends of AI progress, “defence in depth” frameworks for AI safety, and AI governance.
- [Machine Learning Summer School \(MLSS\)](#) (2026): deep learning, probabilistic modelling, trustworthy ML (MPI for Intelligent Systems).
- [Leading for Tomorrow](#) (2025): DPG leadership program on project management, communication, and team building.
- [CERN School of Computing](#) (2024): 50+ hours on software development, machine learning, security and data management.
- [Fast and Efficient Python Programming School](#) (2024): code acceleration, profiling, and GPU programming.

Responsibility, Funding & Awards

- **ErUM-Data Project: DEMOS – 300 000 €** (2025): main author of the successful proposal (Federal Ministry of Research, Technology and Space).
- **Belle II Early Career Researchers Representative** (2024 – 2025): elected by peers, 1000+ member collaboration.
- **Climate Fresk Facilitator** (2022 – present): volunteer facilitator for IPCC-based climate workshops.
- **Franz Mandl Prize** (June 2018): best BSc theoretical physics graduate, University of Manchester.

Selected Publications

- [1] **Lorenz Gärtner**, Nikolai Krug, Thomas Kuhr, Michael A. Schmidt, et al. *Constraints on invisible $B^+ \rightarrow K^+ X$ decays from the Belle II $B^+ \rightarrow K^+ \nu \bar{\nu}$ measurement*. In: *Phys. Rev. D* 114.3 (2026), p. 032003.
- [2] Merna Abumusabh et al. *Model-agnostic likelihood for the reinterpretation of the $B^+ \rightarrow K^+ \nu \bar{\nu}$ measurement at Belle II*. In: *Phys. Rev. D* 112.9 (2025), p. 092016.
- [3] **Lorenz Gärtner**, Nikolai Hartmann, Lukas Heinrich, Malin Horstmann, et al. *Constructing model-agnostic likelihoods, a method for the reinterpretation of particle physics results*. In: *Eur. Phys. J. C* 84.7 (2024), p. 693.
- [4] Johannes Albrecht, Florian Bernlochner, Marco Colonna, **Lorenz Gärtner**, et al. *Joint LHCb–Belle II prospects to constrain new physics in $B \rightarrow D^{(*)} \tau \nu$* . In: *Eur. Phys. J. C* 86.6 (2026), p. 718.
- [5] Jonathan Butterworth et al. *Reinterpretation and preservation of data and analyses in HEP*. Community report. LHC Reinterpretation Forum, CERN-LPCC-2025-002, Mar. 2025.

[2] and [5] carry collaboration-wide author lists; [4] is alphabetical.